LEHMER'S TOTIENT PROBLEM FOR IDEALS OF DEDEKIND DOMAINS

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Abstract. For an ideal I of a Dedekind domain with finite residue fields, Lehmer's question asks when $\varphi(I) = |(R/I)^\times|$ divides $\mathrm{N}(I)-1$ without I being prime. We show that every finite multiset of prime powers is the norm multiset of an ideal in infinitely many number fields, and that for each number of prime factors there are finitely many solutions, uniformly over all Dedekind domains. We solve the problem, and its companion $\varphi(I) \mid \mathrm{N}(I)+1$, for ideals above one odd prime and in odd positive characteristic, solve the companion in characteristic two, and classify the Lehmer ideals of characteristic two with at most twelve primes of norm two. We also list the solutions with few prime factors in quadratic, Gaussian and Eisenstein arithmetic.

1. Introduction

In 1932 D. H. Lehmer asked whether a positive integer n with $\varphi(n) \mid n-1$ must be prime [12]. The question is still open. Every composite solution is odd, squarefree and a Carmichael number, it has at least fifteen prime factors, and with k prime factors it is smaller than k^{2^k} [12, 7, 15, 16]. The difficulty is that the condition mixes the multiplicative structure of n with the additive shift by one, and nothing in the arithmetic of $\mathbb{Z}$ seems to make the shift harmless.

The totient has a natural meaning in any ring: $\varphi(n)$ is the order of the unit group of $\mathbb{Z}/n\mathbb{Z}$. For a Dedekind domain R whose residue fields are finite, and a nonzero ideal I, one has the norm $\mathrm{N}(I) = |R/I|$ and the totient $\varphi(I) = |(R/I)^\times|$, and Lehmer's condition becomes

$$\varphi(I) \mid \mathrm{N}(I) - 1.$$

Ji and Qin [10] solved this problem for $R = \mathbb{F}_q[x]$: there are no solutions for $q \geq 4$, and an explicit short list for $q = 2, 3$. For the Gaussian integers, Grau, Oller-Marcén, Rodríguez and Sadornil studied a different analogue in which the modulus is a rational integer and the group is the norm-one subgroup [9], and Smpokos considered rational integers d in rings of integers with unique factorization, comparing $|(\mathcal{O}_K/d\mathcal{O}_K)^\times|$ with $d^{[K:\mathbb{Q}]} - 1$ [17]. In this paper we study the condition for all ideals of all such Dedekind domains, together with the companion condition $\varphi(I) \mid \mathrm{N}(I)+1$, which Lehmer also considered for integers.

Two kinds of trivial solutions have to be set aside. Prime ideals satisfy the condition, and so does any product of distinct primes of norm 2, because its totient is 1; over $\mathbb{Z}$ the latter is only the prime 2, but a number field in which 2 splits has many. We call an ideal a *Lehmer ideal* if it is not prime, has $\varphi(I) > 1$ and satisfies $\varphi(I) \mid \mathrm{N}(I) - 1$, and a *companion ideal* if the same holds with $\mathrm{N}(I)+1$ (Definition 2.1).

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1.1. **From ideals to multisets.** A Lehmer or companion ideal is squarefree (Lemma 2.2), so only the multiset of the norms of its prime factors matters. A multiset $\mathcal{Q}$ of prime powers is called admissible if $\prod_{q\in\mathcal{Q}}(q-1)$ divides $\prod_{q\in\mathcal{Q}} q\mp 1$ (with at least two members and $\prod(q-1)>1$). Our first result says that number fields impose no restriction on which multisets occur.

Theorem A (Theorem 2.5, Corollary 2.7)**.** *Every finite multiset of prime powers is the multiset of norms of the prime factors of a squarefree ideal, in infinitely many number fields. Hence a multiset is admissible if and only if it is the norm multiset of a Lehmer (or companion) ideal in some number field.*

The proof is local: a polynomial is chosen p-adically close to a product of minimal polynomials of unramified extensions, so that Krasner's lemma fixes the splitting of each relevant prime, and Eisenstein at an auxiliary prime so that it is irreducible.

The second result is the ideal form of the classical finiteness theorem.

Theorem B (Theorem 3.3)**.** *For each k there are finitely many admissible multisets with k members, and each has $\prod q \le (k2^{k+1})^{2^k-1}$. In particular, in every Dedekind domain with finite residue fields a Lehmer or companion ideal with k prime factors has norm at most $(k2^{k+1})^{2^k-1}$.*

1.2. **One characteristic.** An admissible multiset consists either of powers of 2 only or of odd prime powers only (Lemma 2.2). When all members are powers of a single prime, the problem can be solved with Zsigmondy's theorem on primitive divisors and the lifting-the-exponent lemma.

Theorem C (Theorems 5.1, 4.7, 4.3, 4.4)**.** *Let p be a prime and let $\mathcal{Q}$ be a multiset of powers of p.*

(i) *If p is odd, $\mathcal{Q}$ is Lehmer only for $\{3,3\}$ and $\{3,3,3,3\}$, and never companion.*
(ii) *If $p=2$, $\mathcal{Q}$ is companion if and only if it consists of t copies of 2 and $c\ge 1$ copies of 4 with t odd and $3^{c-1}\mid t+2c$.*
(iii) *If $p=2$, for each t there are finitely many Lehmer multisets with t members equal to 2; there are none for $t=0$, only $\{2,4,8\}$ for $t=1$, and 48 in all for $t\le 12$.*

In positive characteristic all residue fields have the same characteristic, so Theorem C answers the question for every Dedekind domain of odd characteristic and for the companion condition in characteristic two, for instance for the coordinate rings of all affine curves over finite fields (Theorem 6.1). For $\mathbb{F}_q[x]$ this recovers the theorem of Ji and Qin and adds its companion form: $\varphi(f)\mid q^{\deg f}+1$ with f reducible and $\varphi(f)>1$ happens only for $q=2$ and $f=x(x^2+x+1)$ or $(x+1)(x^2+x+1)$ (Corollary 6.2).

Theorem D (Theorem 6.3, Corollaries 6.4, 6.5)**.** *Let K be a number field and p a prime. An ideal of $\mathcal{O}_K$ supported above $p\ge 5$ is never a Lehmer or companion ideal. One supported above 3 is a Lehmer ideal exactly when it is a product of two or four distinct primes of residue degree one, and never a companion ideal. A companion ideal supported above 2 exists exactly when 2 has primes of residue degrees 1 and 2. In a Galois number field every Lehmer ideal supported above one prime lies above 3, and a cyclotomic field has none.*

For example $3\mathcal{O}_K$ is a Lehmer ideal of $\mathbb{Q}(\sqrt{d})$ exactly when $d\equiv 1 \pmod 3$, as in $\mathbb{Z}[\sqrt{-2}]$, where $\varphi(3)=4$ divides 3^2-1.

1.3. **Several primes.** When at least two odd primes are involved the problem contains Lehmer's original question, and we can only give lists. Allowing repeated norms and prime powers, as number fields do, produces genuinely new solutions.

Theorem E (Theorems 7.1, 7.4, 8.1, Proposition 7.3). (i) *The Lehmer multisets of odd prime powers with at most seven members are*

$$\{3,3\},\quad \{3,3,3,3\},\quad \{5,5,9,9,89\},\quad \{3,3,5,5,9,89\},\quad \{3,3,3,3,5,5,89\}.$$

The third is realized in the biquadratic field $\mathbb{Q}(i,\sqrt{89})$.

(ii) *In a quadratic field, the only Lehmer ideal with at most 7 prime factors is* $3\mathcal{O}_K$ *(when* 3 *splits). The Gaussian integers have no Lehmer element with at most 10 prime factors and the Eisenstein integers none with at most 14.*

(iii) *No composite integer with at most* 14 *prime factors satisfies* $\varphi(n) \mid n-1$*, and the composite solutions of* $\varphi(n) \mid n+1$ *with at most 7 prime factors are the six known ones. A composite* $n < 10^{37}$ *with* $\varphi(n) \mid n-1$ *would have exactly* 15 *prime factors and satisfy* $n - 1 = 2\varphi(n)$.

Part (iii) reproduces known bounds [7, 16] with a program that is the same for all rings, and adds a size bound (Remark 8.2). Section 9 lists the related conjectures and questions with their status.

1.4. **Organization and verification.** Section 2 sets up ideal totients and proves Theorem A; Section 3 proves Theorem B and describes the search tree. Sections 4 and 5 treat one characteristic, and Section 6 draws the consequences for rings. Section 7 treats several odd primes and Section 8 the integers. Every statement is proved in the text. The finite case checks are done by two independent programs, one in C with GMP and one in Julia, whose outputs agree, and the identities used in the proofs are certified by a Lean 4 file checked by the kernel (Appendices A and B).

2. Ideal totients and norm multisets

2.1. **The totient of an ideal.** Throughout, R is a Dedekind domain such that $R/\mathfrak{p}$ is finite for every nonzero prime ideal $\mathfrak{p}$. The *norm* of a nonzero ideal I is $\mathrm{N}(I) = |R/I|$ and its *totient* is

$$\varphi(I) = |(R/I)^\times|.$$

If $I = \mathfrak{p}_1^{a_1}\cdots\mathfrak{p}_k^{a_k}$ with distinct $\mathfrak{p}_i$, the Chinese remainder theorem gives $R/I \cong \prod_i R/\mathfrak{p}_i^{a_i}$, and $R/\mathfrak{p}^a$ is a local ring of order $\mathrm{N}(\mathfrak{p})^a$ whose maximal ideal $\mathfrak{p}/\mathfrak{p}^a$ has $\mathrm{N}(\mathfrak{p})^{a-1}$ elements. Hence N and φ are multiplicative and

$$\mathrm{N}(I) = \prod_{i=1}^{k} \mathrm{N}(\mathfrak{p}_i)^{a_i}, \qquad \varphi(I) = \prod_{i=1}^{k} \mathrm{N}(\mathfrak{p}_i)^{a_i-1}(\mathrm{N}(\mathfrak{p}_i)-1). \tag{2.1}$$

For $R = \mathbb{Z}$ and $I = n\mathbb{Z}$ these are n and Euler's $\varphi(n)$. For $R = \mathbb{F}_q[x]$ and $I = (f)$ they are $q^{\deg f}$ and the function-field totient studied in [10].

Every prime $\mathfrak{p}$ satisfies $\varphi(\mathfrak{p}) = \mathrm{N}(\mathfrak{p}) - 1$, so both divisibilities $\varphi(\mathfrak{p}) \mid \mathrm{N}(\mathfrak{p}) - 1$ and, when $\mathrm{N}(\mathfrak{p}) \in \{2,3\}$, $\varphi(\mathfrak{p}) \mid \mathrm{N}(\mathfrak{p}) + 1$ hold for trivial reasons. A second trivial case comes from primes of norm 2: if every prime factor of a squarefree ideal I has norm two then $\varphi(I) = 1$, which divides everything. Over $\mathbb{Z}$ the only such ideal is $2\mathbb{Z}$, but a number field in which 2 splits has many of them.

Definition 2.1. Let $\varepsilon \in \{-1,+1\}$. A nonzero ideal I of R is an *ε-Lehmer ideal* if I is not prime, $\varphi(I) > 1$ and $\varphi(I) \mid \mathrm{N}(I) + \varepsilon$. We call (-1)-Lehmer ideals simply *Lehmer ideals* and $(+1)$-Lehmer ideals *companion ideals*. Ideals with $\varphi(I) = 1$ are called *trivial*.

The case $\varepsilon = +1$ goes back to Lehmer's paper as well [12]: over $\mathbb{Z}$ the condition $\varphi(n) \mid n+1$ holds for $n = 3\cdot 5$, $3\cdot 5\cdot 17$, $3\cdot 5\cdot 17\cdot 257$, $3\cdot 5\cdot 17\cdot 257\cdot 65537$, for $3\cdot 5\cdot 17\cdot 353\cdot 929$ and for $3\cdot 5\cdot 17\cdot 353\cdot 929\cdot 83623937$, and no other composite example is known. We recover these six in Section 8.

2.2. **Squarefreeness, independence and parity.** The following lemma is the ideal form of the elementary facts that open every study of Lehmer's problem [12]. For a prime $\mathfrak{p}$ we write $p(\mathfrak{p})$ for the characteristic of $R/\mathfrak{p}$, so that $\mathrm{N}(\mathfrak{p})$ is a power of $p(\mathfrak{p})$.

Lemma 2.2. *Let I be an ε-Lehmer ideal and $M = (\mathrm{N}(I)+\varepsilon)/\varphi(I)$.*

(a) *$I = \mathfrak{p}_1\cdots\mathfrak{p}_k$ is squarefree and $k \geq 2$.*
(b) *For all $i \neq j$, $p(\mathfrak{p}_i) \nmid \mathrm{N}(\mathfrak{p}_j) - 1$; moreover $p(\mathfrak{p}_i) \nmid M$.*
(c) *Either every $\mathrm{N}(\mathfrak{p}_i)$ is a power of 2, or every $\mathrm{N}(\mathfrak{p}_i)$ is odd.*
(d) *With $q_i = \mathrm{N}(\mathfrak{p}_i)$,*

$$\prod_{i=1}^{k} \frac{q_i}{q_i - 1} = M - \frac{\varepsilon}{\varphi(I)}, \tag{2.2}$$

and $M \geq 2$.

Proof. (a) If $\mathfrak{p}^2 \mid I$ then $\mathrm{N}(\mathfrak{p})$ divides $\varphi(I)$ by (2.1), so $p = p(\mathfrak{p})$ divides both $\varphi(I)$ and $\mathrm{N}(I)$. As $\varphi(I) \mid \mathrm{N}(I)+\varepsilon$, p would divide $\varepsilon = \pm 1$. So I is squarefree; it is not prime, hence $k \geq 2$.

(b) The number $q_j - 1$ divides $\varphi(I)$, which divides $\mathrm{N}(I)+\varepsilon$, a number prime to $p(\mathfrak{p}_i)$ because $p(\mathfrak{p}_i) \mid \mathrm{N}(I)$. The same argument applies to M, a divisor of $\mathrm{N}(I)+\varepsilon$.

(c) If some q_i is odd then $q_i - 1$ is even and divides $\mathrm{N}(I)+\varepsilon$, so $\mathrm{N}(I)$ is odd and no q_j is even.

(d) Divide $M\varphi(I) = \mathrm{N}(I)+\varepsilon$ by $\varphi(I) = \prod(q_i - 1)$. The left side of (2.2) exceeds 1, so $M \geq 1$; if $\varepsilon = +1$ it exceeds $1 + 1/\varphi(I)$ and $M \geq 2$. If $\varepsilon = -1$ and $M = 1$ then $\prod q_i - \prod(q_i - 1) = 1$, but for $k \geq 2$ the left side is at least $q_1 q_2 - (q_1 - 1)(q_2 - 1) = q_1 + q_2 - 1 \geq 3$. □

Part (b) is the analogue of the condition $p \mid n \Rightarrow p \nmid q - 1$ for primes p, q dividing a Lehmer number; it already forces a Lehmer integer to be odd, while for ideals it leaves room for an even case, which turns out to be the richest one (Section 4).

2.3. **Norm multisets.** By Lemma 2.2(a) only the multiset of norms of the prime factors matters.

Definition 2.3. A *norm multiset* is a finite multiset $\mathcal{Q} = \{q_1, \ldots, q_k\}$ of prime powers, $k \geq 1$. We put $\mathrm{N}(\mathcal{Q}) = \prod q_i$ and $\varphi(\mathcal{Q}) = \prod(q_i - 1)$, and call $\mathcal{Q}$ *ε-admissible* if $k \geq 2$, $\varphi(\mathcal{Q}) > 1$ and $\varphi(\mathcal{Q}) \mid \mathrm{N}(\mathcal{Q}) + \varepsilon$. The multiset of norms of the prime factors of a squarefree ideal I is written $\mathcal{Q}(I)$.

Lemma 2.4. *An ideal I is an ε-Lehmer ideal if and only if it is squarefree and $\mathcal{Q}(I)$ is ε-admissible.*

Proof. For squarefree I, (2.1) gives $\mathrm{N}(I) = \mathrm{N}(\mathcal{Q}(I))$ and $\varphi(I) = \varphi(\mathcal{Q}(I))$, and I is prime exactly when $|\mathcal{Q}(I)| = 1$. Combine with Lemma 2.2(a). □

Lemma 2.2(b)–(d) are statements about admissible multisets; we use them in that form. In particular, an admissible multiset satisfies the *independence condition*: if q and q' are two of its members (possibly equal as numbers but distinct as members) and p is the prime of which q is a power, then $p \nmid q' - 1$.

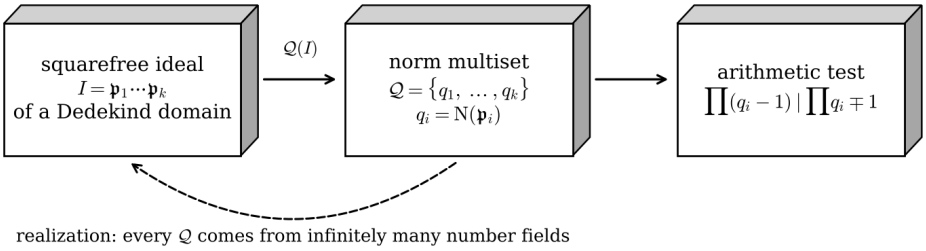


FIGURE 1. The reduction of Lemma 2.4 and Theorem 2.5. An ideal is sent to the multiset of the norms of its prime factors; every multiset arises from ideals of infinitely many number fields, and the arithmetic of the ring only restricts which multisets occur.

2.4. **Every multiset occurs in number fields.** Whether a given multiset occurs as $\mathcal{Q}(I)$ depends on the ring: in $\mathbb{Z}$ only multisets of distinct primes occur, and in a quadratic field a prime p contributes at most two norms. Over all number fields together there is no restriction.

Theorem 2.5. *Let $\mathcal{Q}$ be a norm multiset. There are infinitely many number fields K such that $\mathcal{O}_K$ has a squarefree ideal I with $\mathcal{Q}(I) = \mathcal{Q}$. If $p_1, \ldots, p_s$ are the primes dividing the members of $\mathcal{Q}$ and $\mathcal{Q}_j = \{p_j^{f_{j,1}}, \ldots, p_j^{f_{j,r_j}}\}$ is the part of $\mathcal{Q}$ above p_j, the fields can be chosen of any degree $n \geq \max_j(f_{j,1} + \cdots + f_{j,r_j})$.*

Proof. Fix n as in the statement. For each j let $m_j = n - \sum_i f_{j,i} \geq 0$. For each i let $L_{j,i}$ be the unramified extension of $\mathbb{Q}_{p_j}$ of degree $f_{j,i}$; it is generated by a root of unity $\zeta_{j,i}$ of order $p_j^{f_{j,i}} - 1$, and also by every element $\alpha_{j,i} = \zeta_{j,i} + c_{j,i}p_j$ with $c_{j,i} \in \mathbb{Z}$, because the residue class of $\alpha_{j,i}$ generates the residue field $\mathbb{F}_{p_j^{f_{j,i}}}$. Such an element is a unit of $L_{j,i}$. Let $G_{j,i} \in \mathbb{Z}_{p_j}[x]$ be its minimal polynomial. If two of these elements, $\zeta + cp_j$ and $\zeta' + c'p_j$, are conjugate, then $\zeta - \sigma(\zeta') = (c' - c)p_j$ for some automorphism σ; a difference of two distinct roots of unity of order prime to p_j is a unit, so $\zeta = \sigma(\zeta')$ and $c = c'$. Choosing the integers $c_{j,i}$ distinct we therefore make $G_{j,1}, \ldots, G_{j,r_j}$ pairwise distinct; being irreducible, they are then pairwise coprime. If $m_j \geq 1$ put $H_j = x^{m_j} - p_j$, and otherwise $H_j = 1$. The roots of H_j have positive valuation and the roots of the $G_{j,i}$ are units, so H_j is coprime to every $G_{j,i}$. Let

$$G_j = H_j \prod_{i=1}^{r_j} G_{j,i} \in \mathbb{Z}_{p_j}[x],$$

a monic separable polynomial of degree n with $\mathbb{Q}_{p_j}[x]/(G_j) \cong \mathbb{Q}_{p_j}[x]/(H_j) \times \prod_i L_{j,i}$.

Choose a prime ℓ different from all p_j. By the Chinese remainder theorem applied to each coefficient, there are monic $g \in \mathbb{Z}[x]$ of degree n with $g \equiv x^n \pmod{\ell}$, $g(0) \equiv \ell \pmod{\ell^2}$, and $g \equiv G_j \pmod{p_j^N}$ for each j, where N is any fixed large integer. Such a g is Eisenstein at ℓ, hence irreducible over $\mathbb{Q}$, and $K = \mathbb{Q}[x]/(g)$ is a field of degree n in which ℓ is totally ramified.

We show that for N large, $\mathbb{Q}_{p_j}[x]/(g) \cong \mathbb{Q}_{p_j}[x]/(G_j)$. Write $p = p_j$ and let $\delta > 0$ be smaller than every distance $|\alpha - \alpha'|_p$ between distinct roots of the separable polynomial G_j in an algebraic closure of $\mathbb{Q}_p$. Roots depend continuously on the coefficients of a monic polynomial, so for N large every root of g lies within $\delta/2$ of a root of G_j, and each root α of G_j has exactly

one root β_α of g within $\delta/2$ of it. Then $|\beta_\alpha - \alpha|_p < |\alpha - \alpha'|_p$ for every other root α' of G_j, so $\mathbb{Q}_p(\alpha) \subseteq \mathbb{Q}_p(\beta_\alpha)$ by Krasner's lemma [11, Ch. II, §2], and symmetrically $|\alpha - \beta_\alpha|_p < |\beta_\alpha - \beta'|_p$ for every other root β' of g, so $\mathbb{Q}_p(\beta_\alpha) \subseteq \mathbb{Q}_p(\alpha)$. The bijection $\alpha \mapsto \beta_\alpha$ commutes with the action of the absolute Galois group of $\mathbb{Q}_p$, since that action preserves distances. Hence the irreducible factors of g over $\mathbb{Q}_p$ correspond to those of G_j, with the same root fields. Therefore

$$K \otimes_{\mathbb{Q}} \mathbb{Q}_{p_j} \cong \mathbb{Q}_{p_j}[x]/(H_j) \times \prod_{i=1}^{r_j} L_{j,i}.$$

The primes of $\mathcal{O}_K$ above p_j correspond to the field factors of $K \otimes \mathbb{Q}_{p_j}$, and the residue degree of each is the residue degree of its factor [14, Ch. II, §8]. So for each i there is a prime $\mathfrak{p}_{j,i}$ of $\mathcal{O}_K$ above p_j with $\mathrm{N}(\mathfrak{p}_{j,i}) = p_j^{f_{j,i}}$, and these primes are distinct for distinct i. The ideal $I = \prod_{j,i} \mathfrak{p}_{j,i}$ is squarefree with $\mathcal{Q}(I) = \mathcal{Q}$.

Finally, a number field has finitely many ramified primes, while ℓ ramifies in K. Letting ℓ run through infinitely many primes gives infinitely many distinct fields K. □

Remark 2.6. The construction gives degree $n = \max_j \sum_i f_{j,i}$, which cannot be lowered: the residue degrees of the primes above p_j that divide I add up to at most $[K : \mathbb{Q}]$. When the number of equal members of $\mathcal{Q}$ above p does not exceed the number of monic irreducible polynomials of the corresponding degree over $\mathbb{F}_p$, one can avoid Krasner's lemma and take a polynomial whose reduction modulo p is a product of distinct irreducible factors of the right degrees; the local construction above is needed only when a norm occurs more often than that, for example for six primes of norm 5.

Corollary 2.7. *A norm multiset $\mathcal{Q}$ is ε-admissible if and only if there is a number field with an ε-Lehmer ideal I such that $\mathcal{Q}(I) = \mathcal{Q}$; in that case there are infinitely many such fields.*

Proof. Lemma 2.4 and Theorem 2.5. □

So Lehmer's problem over all number fields at once is exactly the problem of describing the admissible multisets. Rings with more structure, such as $\mathbb{Z}$, $\mathbb{Z}[i]$ or the rings of integers of quadratic fields, add conditions on which multisets may occur, and positive characteristic forces all members of $\mathcal{Q}$ to be powers of one prime. The rest of the paper studies admissible multisets under these constraints.

3. Finiteness for a fixed number of prime factors

For integers, Pomerance [15] proved that a Lehmer number with k prime factors is smaller than k^{2^k}. The argument behind such bounds uses only the product formula (2.2), so it works for multisets of prime powers and gives a bound that is uniform over all Dedekind domains. We present it in the form in which it also drives our searches.

3.1. **The bound.** Let $\mathcal{Q} = \{q_1 \le q_2 \le \cdots \le q_k\}$ be ε-admissible, with M as in Lemma 2.2. For $0 \le j \le k$ put

$$A_j = \prod_{i \le j} q_i, \qquad B_j = \prod_{i \le j} (q_i - 1), \qquad C_j = MB_j - A_j,$$

so $A_0 = B_0 = 1$.

Lemma 3.1. *$M \le 2^k$, and $C_j \ge 1$ for $0 \le j \le k-1$.*

Proof. Each factor $q_i/(q_i-1)$ is at most 2, with equality only for $q_i = 2$. If $\varepsilon = -1$ then $M < \mathrm{N}(\mathcal{Q})/\varphi(\mathcal{Q}) \le 2^k$ by (2.2). If $\varepsilon = +1$ then $\varphi(\mathcal{Q}) \ge 2$ means some $q_i \ge 3$, so $\mathrm{N}/\varphi \le 2^{k-1}\cdot\frac{3}{2}$ and $M = \mathrm{N}/\varphi + 1/\varphi < 2^k$.

For C_j, note $A_j/B_j \le A_{k-1}/B_{k-1}$. If $\varepsilon = +1$ then $A_{k-1}/B_{k-1} \le A_k/B_k = M - 1/\varphi < M$. If $\varepsilon = -1$ then $A_{k-1}/B_{k-1} = (M + 1/\varphi)(1 - 1/q_k)$, which is less than M exactly when $q_k < M\varphi + 1 = \mathrm{N}(\mathcal{Q})$, and this holds because $k \ge 2$. So $MB_j - A_j > 0$. □

Proposition 3.2. *Let $0 \le j \le k-1$ and $r = k - j$. Then*

$$\frac{MB_j}{C_j} < q_{j+1} \le \frac{2rMB_j}{C_j} \quad (j \le k-2), \qquad q_k = \frac{MB_{k-1} + \varepsilon}{C_{k-1}}.$$

For $\varepsilon = -1$ the factor 2 in the upper bound can be omitted. Moreover q_{j+1} satisfies the reach condition

$$A_j q_{j+1}^r > MB_j(q_{j+1}-1)^r \quad (\varepsilon = -1), \qquad A_j q_{j+1}^r + 1 \ge MB_j(q_{j+1}-1)^r \quad (\varepsilon = +1). \tag{3.1}$$

Proof. For the lower bound, $j + 1 \le k - 1$ and Lemma 3.1 give $A_j q_{j+1} < MB_j(q_{j+1} - 1)$, that is $q_{j+1}C_j > MB_j$. The formula for q_k is $MB_{k-1}(q_k - 1) = A_{k-1}q_k + \varepsilon$ solved for q_k.

For the reach condition, the members after q_j are at least $q = q_{j+1}$, so $\prod_{i>j} q_i/(q_i-1) \le (q/(q-1))^r$ and $\prod_{i>j}(q_i - 1) \ge (q-1)^r$. By (2.2), $\prod_{i>j} q_i/(q_i-1) = (M - \varepsilon/\varphi)B_j/A_j$ and $\varphi = B_j \prod_{i>j}(q_i - 1)$. For $\varepsilon = -1$ this gives $(q/(q-1))^r > MB_j/A_j$; for $\varepsilon = +1$ it gives $(q/(q-1))^r \ge MB_j/A_j - 1/(A_j(q-1)^r)$. Clearing denominators yields (3.1).

For the upper bound we may assume $q > r$. Bernoulli's inequality gives $(1-1/q)^r \ge 1 - r/q$, so $(q/(q-1))^r \le (1 - r/q)^{-1}$. If $\varepsilon = -1$, the inequality $(1 - r/q)^{-1} > MB_j/A_j$ rearranges to $r/q > C_j/(MB_j)$, that is $q < rMB_j/C_j$. If $\varepsilon = +1$ we have $(1 - r/q)^{-1} \ge (MB_j - B_j/\varphi)/A_j$ with $B_j/\varphi = 1/\prod_{i>j}(q_i - 1)$. If every q_i with $i > j$ equals 2, then $q = 2$ and the bound holds. Otherwise $B_j/\varphi \le \frac{1}{2} \le C_j/2$, so $(1 - r/q)^{-1} \ge (MB_j - B_j/\varphi)/A_j \ge (A_j + C_j/2)/A_j$, which rearranges to $q \le 2r(A_j + C_j/2)/C_j \le 2rMB_j/C_j$. □

Theorem 3.3. *For every $k \ge 2$ and $\varepsilon \in \{\pm 1\}$ there are finitely many ε-admissible multisets with k members, and each satisfies*

$$\mathrm{N}(\mathcal{Q}) \le \left(k\, 2^{k+1}\right)^{2^k - 1}.$$

Consequently, in any Dedekind domain with finite residue fields, an ε-Lehmer ideal with k prime factors has norm at most $(k2^{k+1})^{2^k-1}$, and only finitely many norm multisets occur among such ideals, for all such domains together.

Proof. Put $T = k2^{k+1}$. By Proposition 3.2 and Lemma 3.1, $q_{j+1} \le 2kMB_j \le TB_j < TA_j$ for $j \ge 1$, and $q_1 \le 2kM \le T$; the case $j = k - 1$ follows from the formula for q_k, since $MB_{k-1} + \varepsilon \le TB_{k-1}$. Hence $A_1 \le T$ and $A_{j+1} \le TA_j^2$, and induction gives $A_j \le T^{2^j - 1}$. For $j = k$ this is the claim. The second statement follows from Lemma 2.4. □

The bound is far from the truth, and it is not what we use in computation. What matters is that Proposition 3.2 turns the search for admissible multisets into a finite tree.

3.2. **The search tree.** Fix k, ε and M. A *node* at depth j is a nondecreasing sequence $q_1 \le \cdots \le q_j$ of prime powers satisfying the independence condition of Lemma 2.2(b), with $C_j \ge 1$ and $p(q_i) \nmid M$. Its children are the extensions by a prime power $q \ge q_j$ in the interval

$$\max\Big(q_j, \Big\lfloor \frac{MB_j}{C_j} \Big\rfloor + 1\Big) \le q \le \max\{q : q \text{ satisfies (3.1) with } r = k - j\}.$$

The right end is found by bisection, because (3.1) fails for all large q once it fails for one q (the ratio $q^r/(q-1)^r$ decreases). At depth $k-2$ we do not branch twice. For each admissible $q = q_{k-1}$ the last member is forced: putting $A = A_{k-2}q$, $B = B_{k-2}(q-1)$, $C = MB - A$, it must equal $x = (MB+\varepsilon)/C$, and we only test whether x is an integer prime power at least q that is compatible with the earlier members. Equivalently, the last two members solve

$$(C'q - MB')(C'x - MB') = M^2B'^2 - C'(MB' - \varepsilon), \tag{3.2}$$

where $A' = A_{k-2}$, $B' = B_{k-2}$ and $C' = MB' - A'$; this is how one checks by hand that a terminal node has no completion.

Different rings impose different rules on the multisets, all of which are local conditions on the members and are applied at every node:

- Z (rational integers) the members are distinct primes;
- U (all number fields) the members are arbitrary prime powers with repetition;
- Q (quadratic fields) each prime p contributes at most two members, and they are $\{p\}$, $\{p,p\}$ or $\{p^2\}$;
- G (the Gaussian integers $\mathbb{Z}[i]$) the members above $p \equiv 1 \pmod 4$ are at most two copies of p, a prime $p \equiv 3 \pmod 4$ contributes at most one p^2, and 2 contributes at most one member 2;
- E (the Eisenstein integers $\mathbb{Z}[\omega]$) the members above $p \equiv 1 \pmod 3$ are at most two copies of p, a prime $p \equiv 2 \pmod 3$ contributes at most one p^2, and 3 contributes at most one member 3.

In Section 4 we show that admissible multisets of powers of 2 never mix with odd members, so in the modes Q, G, E and U the searches are run over odd prime powers only and the even case is settled separately. In mode Z we add one more rule that removes most of the work for small M. If $q_1 = 3$, every other member is a prime $q \equiv 2 \pmod 3$ by independence, so $\varphi(\mathcal{Q}) \equiv 2$ and $\mathrm{N}(\mathcal{Q}) \equiv 0 \pmod 3$; then $M\varphi(\mathcal{Q}) = \mathrm{N}(\mathcal{Q}) + \varepsilon$ gives $2M \equiv \varepsilon$, that is $M \equiv 1 \pmod 3$ for $\varepsilon = -1$ and $M \equiv 2 \pmod 3$ for $\varepsilon = +1$.

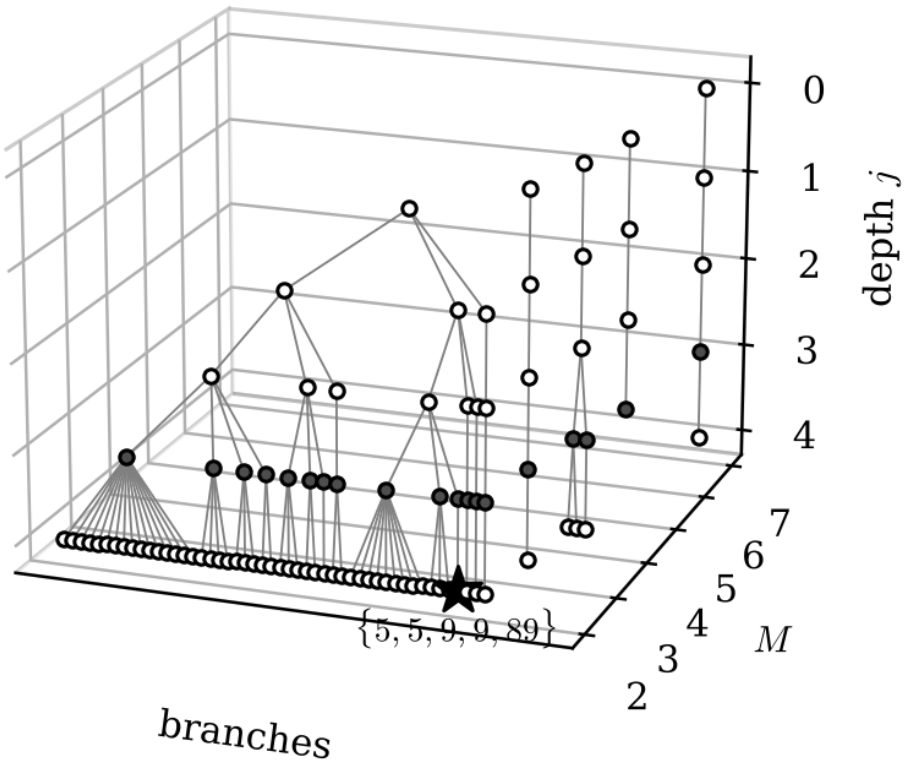


FIGURE 2. The search tree of Section 3.2 for $k = 5$, $\varepsilon = -1$ in mode U, drawn by depth. Each node is a sorted prefix; its children fill the interval of Proposition 3.2, and at depth $k-2$ the last member is forced. The highlighted branch ends in the admissible multiset $\{5, 5, 9, 9, 89\}$; all other leaves are dead.

4. Characteristic two

By Lemma 2.2(c) an admissible multiset consists either of powers of 2 or of odd prime powers. This section treats the first kind. We write a multiset of powers of 2 as

$$\mathcal{Q} = \{\underbrace{2, \dots, 2}_{t}\} \cup \{2^f : f \in S\},$$

where S is a multiset of integers $f \geq 2$, and put $F = t + \sum_{f \in S} f$, so $\mathrm{N}(\mathcal{Q}) = 2^F$ and $\varphi(\mathcal{Q}) = \prod_{f \in S}(2^f - 1)$. The condition $\varphi(\mathcal{Q}) > 1$ says $S \neq \emptyset$. Thus $\mathcal{Q}$ is Lehmer if and only if

$$\prod_{f \in S} (2^f - 1) \;\Big|\; 2^F - 1, \tag{4.1}$$

and companion if and only if the same product divides $2^F + 1$.

We use two classical tools. For $n \geq 1$ let Φ_n be the n-th cyclotomic polynomial. A prime ℓ is a *primitive divisor* of $a^n - 1$ if $\ell \mid a^n - 1$ and $\ell \nmid a^m - 1$ for $m < n$.

Lemma 4.1 (Bang, Zsigmondy)**.** *Let $a \geq 2$ and $n \geq 2$. Then $a^n - 1$ has a primitive divisor, except when $(a, n) = (2, 6)$, or $n = 2$ and $a+1$ is a power of 2. The primitive divisors of $a^n - 1$ divide $\Phi_n(a)$, and if $n \geq 3$ then $\Phi_n(a) = P_n^*(a)\, r$, where $P_n^*(a)$ is the part of $\Phi_n(a)$ composed of primitive divisors and r is 1 or the largest prime factor of n.*

This is in [2, 18]; the description of $\Phi_n(a)/P_n^*(a)$ is standard (see for example [4, Section 2]). We also use the lifting-the-exponent lemma: if ℓ is an odd prime not dividing a, o is the order of a modulo ℓ and $o \mid n$, then

$$\nu_\ell(a^n - 1) = \nu_\ell(a^o - 1) + \nu_\ell(n/o), \tag{4.2}$$

while $\ell \nmid a^n - 1$ if $o \nmid n$. Finally, for $n \geq 1$,

$$\varphi(n) \geq \sqrt{n/2}, \tag{4.3}$$

since $p^{a-1}(p-1) \geq p^{a/2}$ for odd p and $2^{a-1} \geq 2^{a/2}/\sqrt{2}$.

4.1. **Lehmer multisets of powers of two.**

Lemma 4.2. *Let $\mathcal{Q}$ be a Lehmer multiset of powers of 2 with t members equal to 2.*

(a) *Every $f \in S$ divides F, and $f < F$.*
(b) *If $f \in S$ occurs m times in S, then $m \leq 1 + \nu_\ell(F)$ for every odd prime ℓ dividing $2^f - 1$; in particular $m \leq 1 + \log_3 F$.*

Proof. (a) $2^f - 1 \mid 2^F - 1$ forces $f \mid F$. If $f = F$ then $S = \{F\}$ and $t = 0$, so $\mathcal{Q}$ has one member, which is excluded.

(b) Let o be the order of 2 modulo ℓ, so $o \mid f \mid F$. By (4.2), $m\,\nu_\ell(2^f - 1) \leq \nu_\ell(2^F - 1) = \nu_\ell(2^o - 1) + \nu_\ell(F/o)$, and $\nu_\ell(2^f - 1) \geq \nu_\ell(2^o - 1) \geq 1$, so $m \leq 1 + \nu_\ell(F)$. Since $\ell \geq 3$, $\nu_\ell(F) \leq \log_3 F$. □

Put

$$c_0 = \prod_{d \geq 1} (1 - 2^{-d}) = 0.288788\ldots, \qquad C_2 = \prod_{v \geq 2} (1 - 2^{-v})^{-1} = 1.731372\ldots.$$

Theorem 4.3. *Let $t \geq 0$. If $\mathcal{Q}$ is a Lehmer multiset of powers of 2 with t members equal to 2, then F lies in the finite set*

$$E_t = \Big\{ F \geq 3 :\ c_0\, 2^{\varphi(F)}/F \leq 2^t\, C_2^{\,1+\log_3 F} \Big\}.$$

In particular, for each t there are finitely many such multisets, and they can be listed by running through the multisets S of proper divisors $f \geq 2$ of $F \in E_t$ with $\sum_{f \in S} f = F - t$.

Proof. $S \neq \emptyset$ and $k \geq 2$ give $F \geq 3$. Let $M = (2^F - 1)/\varphi(\mathcal{Q})$. By Lemma 4.2(a) every $f \in S$ is smaller than F, so no primitive divisor of $2^F - 1$ divides $\varphi(\mathcal{Q})$, and the whole primitive part $P_F^*(2)$ divides M. By Lemma 4.1, $P_F^*(2) \geq \Phi_F(2)/F$. Moreover

$$\Phi_F(2) = \prod_{d \mid F} (2^d - 1)^{\mu(F/d)} = 2^{\varphi(F)} \prod_{d \mid F} (1 - 2^{-d})^{\mu(F/d)} \geq 2^{\varphi(F)} \prod_{d \geq 1} (1 - 2^{-d}),$$

because the factors with $\mu(F/d) = -1$ are at least 1 and those with $\mu(F/d) = +1$ are at most 1 and pairwise distinct. So $M \geq c_0\, 2^{\varphi(F)}/F$.

On the other hand $M < \mathrm{N}(\mathcal{Q})/\varphi(\mathcal{Q}) = 2^t \prod_{f \in S} (1 - 2^{-f})^{-1}$, and by Lemma 4.2(b) each $f \geq 2$ occurs at most $1 + \log_3 F$ times, so $M < 2^t C_2^{\,1+\log_3 F}$. Hence $F \in E_t$. By (4.3) the left side of the defining inequality of E_t is at least $c_0\, 2^{\sqrt{F/2}}/F$, which grows faster than any power of F, so E_t is finite. □

For $t \le 12$ the sets E_t are small: $\max E_t$ is 30 for $t \le 1$, 42 for $2 \le t \le 4$, 60 for $5 \le t \le 8$, 66 for $9 \le t \le 11$ and 90 for $t = 12$, and $|E_t| \le 44$. For $F \ge 2000$ the bound $\varphi(F) \ge \sqrt{F/2}$ already excludes F for all $t \le 12$, and below 2000 we use the exact value of $\varphi(F)$. Running through the divisor multisets gives at most 22183 candidates for each t.

Theorem 4.4. *There is no Lehmer multiset of powers of* 2 *with* $t = 0$*, the only one with* $t = 1$ *is* $\{2, 4, 8\}$*, and the complete list for* $t \le 12$ *is Table 1:* 48 *multisets in all.*

Proof. By Theorem 4.3 it suffices to test (4.1) for the candidates just described; this is a finite computation, done independently by the Python and Julia programs of Appendix A, and the divisibilities in Table 1 are certified in Lean (Appendix B). For $t \le 1$ the candidates are few enough to check by hand from the factorizations of $2^F - 1$, $F \le 30$. □

Table 1: All Lehmer multisets of powers of two with $t \le 12$ members equal to 2. The column S lists the exponents $f \ge 2$ of the other members 2^f; $F = t + \sum_{f \in S} f$ and $\mathrm{N}(\mathcal{Q}) = 2^F$; $M = (2^F - 1)/\varphi(\mathcal{Q})$; $k = t + |S|$ is the number of members.

t	S	F	M	k
1	$\{3, 2\}$	6	3	3
2	$\{2\}$	4	5	3
2	$\{2, 2\}$	6	7	4
3	$\{3\}$	6	9	4
3	$\{5, 2\}$	10	11	5
3	$\{4, 3, 2\}$	12	13	6
3	$\{9, 2, 2, 2\}$	18	19	7
4	$\{2\}$	6	21	5
4	$\{4\}$	8	17	5
5	$\{5\}$	10	33	6
5	$\{4, 3\}$	12	39	7
5	$\{3, 2, 2\}$	12	65	8
5	$\{7, 2\}$	14	43	7
5	$\{9, 2, 2\}$	18	57	8
6	$\{2\}$	8	85	7
6	$\{3\}$	9	73	7
6	$\{6\}$	12	65	7
6	$\{4, 2\}$	12	91	8
7	$\{3, 2\}$	12	195	9
7	$\{7\}$	14	129	8
7	$\{5, 3\}$	15	151	9
7	$\{9, 2\}$	18	171	9
8	$\{2\}$	10	341	9
8	$\{4\}$	12	273	9
8	$\{2, 2\}$	12	455	10
8	$\{8\}$	16	257	9
8	$\{7, 3, 3\}$	21	337	11
9	$\{3\}$	12	585	10
9	$\{9\}$	18	513	10

t	S	F	M	k
9	$\{3,2,2,2\}$	18	1387	13
9	$\{11,2\}$	22	683	11
10	$\{2\}$	12	1365	11
10	$\{5\}$	15	1057	11
10	$\{6,2\}$	18	1387	12
10	$\{10\}$	20	1025	11
11	$\{3,2,2\}$	18	4161	14
11	$\{5,4\}$	20	2255	13
11	$\{7,3\}$	21	2359	13
11	$\{11\}$	22	2049	12
11	$\{8,3,2\}$	24	3133	14
11	$\{13,2\}$	26	2731	13
11	$\{15,2,2\}$	30	3641	14
12	$\{2\}$	14	5461	13
12	$\{3\}$	15	4681	13
12	$\{4\}$	16	4369	13
12	$\{6\}$	18	4161	13
12	$\{2,2,2\}$	18	9709	15
12	$\{12\}$	24	4097	13

Several infinite families are visible in the table.

Proposition 4.5. *Let $t \geq 1$ and $c \geq 1$.*

(a) *For every divisor $f \geq 2$ of t, the multiset $\{2^{(t)}, 2^f\}$ (t copies of 2 and one 2^f) is Lehmer.*
(b) *The multiset $\{2^{(t)}, 4^{(c)}\}$ is Lehmer if and only if t is even and $3^{c-1} \mid t+2c$.*

Proof. (a) $2^f - 1 \mid 2^{t+f} - 1$ because $f \mid t+f$. (b) Here $F = t+2c$ and the condition is $3^c \mid 2^F - 1$. It requires F even, i.e. t even, and then $\nu_3(2^F - 1) = 1 + \nu_3(F)$ by (4.2) with $o = 2$, which is at least c exactly when $3^{c-1} \mid F$. □

For example $\{2^{(t)}, 4, 4\}$ is Lehmer exactly when $t \equiv 2 \pmod 6$ and $\{2^{(t)}, 4, 4, 4\}$ exactly when $t \equiv 12 \pmod{18}$. Table 1 also shows that every multiset in it with $t \geq 2$ contains an exponent $f \leq t$; we do not know whether this holds for all t.

Question 4.6. *Is it true that every Lehmer multiset of powers of 2 with $t \geq 2$ members equal to 2 has a member 2^f with $2 \leq f \leq t$?*

4.2. **Companion multisets of powers of two.** The companion condition in characteristic two has a complete answer.

Theorem 4.7. *A multiset of powers of 2 is a companion multiset if and only if it has the form $\{2^{(t)}, 4^{(c)}\}$ with t odd, $c \geq 1$ and $3^{c-1} \mid t+2c$.*

Proof. Suppose $\mathcal{Q}$ is companion and $f \in S$ with $f \geq 3$. Then $2^f - 1 \mid 2^F + 1 \mid 2^{2F} - 1$. The order of 2 modulo $2^f - 1$ is f, so $f \mid 2F$; and $f \nmid F$, since otherwise $2^f - 1 \mid 2^F - 1$ and $2^f - 1 \mid 2$. Hence f is even and $F \equiv f/2 \pmod f$, so $2^F + 1 \equiv 2^{f/2} + 1 \pmod{2^f - 1}$. But $0 < 2^{f/2} + 1 < 2^f - 1$ for $f \geq 4$, a contradiction. So S consists of $c \geq 1$ copies of 2, and the condition is $3^c \mid 2^F + 1$ with $F = t + 2c$. Since $2^F + 1 \equiv 0 \pmod 3$ only for odd F, t must be odd, and then $\nu_3(2^F + 1) = \nu_3(2^{2F} - 1) - \nu_3(2^F - 1) = 1 + \nu_3(F)$ by (4.2). This is at least c exactly when $3^{c-1} \mid F$. The converse is the same computation read backwards. □

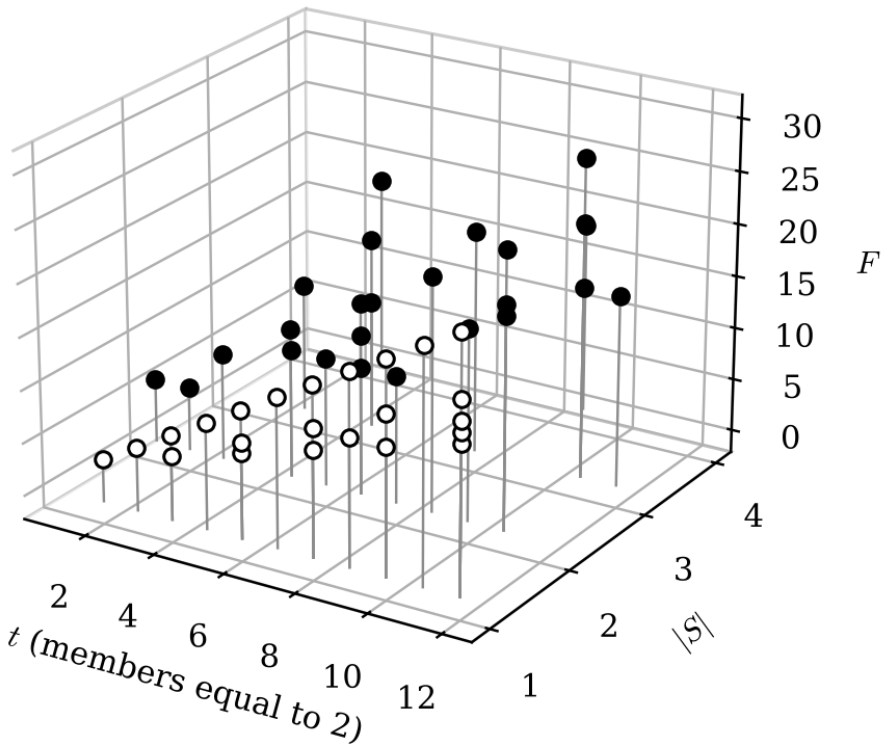


FIGURE 3. Lehmer multisets of powers of two (Table 1), drawn at height F above the point $(t, |S|)$. The diagonal families of Proposition 4.5(a) are the points with $|S| = 1$.

Theorem 4.7 and Proposition 4.5(b) show that the two conditions differ only in the parity of t when only norms 2 and 4 occur: $\{2^{(t)}, 4^{(c)}\}$ is Lehmer for even t and companion for odd t, under the same condition $3^{c-1} \mid t + 2c$.

5. Multisets above one odd prime

Now let p be an odd prime and let every member of $\mathcal{Q}$ be a power of p: $\mathcal{Q} = \{p^f : f \in S\}$ for a multiset S of positive integers with $k = |S| \geq 2$. Put $F = \sum_{f \in S} f$. Then $\mathcal{Q}$ is Lehmer if and only if

$$\prod_{f \in S} (p^f - 1) \;\Big|\; p^F - 1, \tag{5.1}$$

and companion if and only if the product divides $p^F + 1$. (Here $\varphi(\mathcal{Q}) > 1$ holds automatically.)

Theorem 5.1. *Let p be an odd prime.*

(a) *A multiset of powers of p is Lehmer if and only if $p = 3$ and it is $\{3, 3\}$ or $\{3, 3, 3, 3\}$, with $M = 2$ and $M = 5$ respectively.*

(b) *No multiset of powers of p is a companion multiset.*

The two multisets in (a) are Lehmer: $\varphi = 4$ divides $9-1$ and $\varphi = 16$ divides $81-1$. For the proof put

$$c_p = \prod_{d\ge 1}(1-p^{-d}), \qquad c_3 = 0.560126\ldots, \quad c_5 = 0.760332\ldots.$$

Proof of (b). Suppose $\prod_{f\in S}(p^f-1) \mid p^F+1$. If $f \in S$ and $f \ge 2$, the argument in the proof of Theorem 4.7 applies with 2 replaced by p: the order of p modulo p^f-1 is f, so f is even, $F \equiv f/2 \pmod f$, and $p^F+1 \equiv p^{f/2}+1 \pmod{p^f-1}$ with $0 < p^{f/2}+1 < p^f-1$. Hence every f equals 1, $F = k$, and $(p-1)^k \mid p^k+1$. Since $p^k \equiv 1 \pmod{p-1}$, this gives $p-1 \mid 2$, so $p = 3$ and $2^k \mid 3^k+1$. But $\nu_2(3^k+1)$ is 1 for even k and 2 for odd k, so $k \le 1$, a contradiction. □

Proof of (a). Let $\mathcal{Q}$ be Lehmer and $M = (p^F-1)/\varphi(\mathcal{Q})$. As in Lemma 4.2(a), every $f \in S$ divides F and is smaller than F.

The 2-adic bound. Each factor p^f-1 is even, so $k \le \nu_2(p^F-1)$. By (4.2)-type reasoning for the prime 2, $\nu_2(p^F-1)$ equals $\nu_2(p-1)$ for odd F and $\nu_2(p^2-1)+\nu_2(F)-1$ for even F; in both cases

$$2^k \le \frac{(p^2-1)F}{2}. \tag{5.2}$$

The case $F = 2$. Then $S = \{1,1\}$ and (5.1) reads $(p-1)^2 \mid p^2-1$, that is $p-1 \mid p+1$, so $p-1 \mid 2$ and $p = 3$. This gives $\{3,3\}$.

The case $F \ge 3$. By Lemma 4.1, p^F-1 has a primitive divisor; as all $f < F$, the primitive part $P_F^*(p) \ge \Phi_F(p)/F$ divides M. Each complex root ζ of Φ_F satisfies $|p-\zeta| \ge p-1$, so $\Phi_F(p) \ge (p-1)^{\varphi(F)}$, and the argument of Theorem 4.3 also gives $\Phi_F(p) \ge c_p\, p^{\varphi(F)}$. On the other side, $M < \mathrm{N}(\mathcal{Q})/\varphi(\mathcal{Q}) \le (p/(p-1))^k$. With (5.2) we get

$$\frac{\max\big((p-1)^{\varphi(F)},\, c_p\, p^{\varphi(F)}\big)}{F} < \Big(\frac{(p^2-1)F}{2}\Big)^{\log_2(p/(p-1))}. \tag{5.3}$$

Let $p \ge 7$. Then $\log_2(p/(p-1)) \le 1/((p-1)\log 2) \le 0.2405$ and $\log p \le 1.09\log(p-1)$, so the logarithm of the right side of (5.3) is at most $0.525\log(p-1) + 0.2405\log F$. The logarithm of the left side is at least $\varphi(F)\log(p-1) - \log F$. Since $\varphi(F) \ge 2$ and $\log(p-1) \ge \log 6 > 1.7917$, (5.3) would give

$$g(F) := 1.7917\,(\varphi(F) - 0.525) - 1.2405\log F < 0.$$

For $3 \le F \le 7$ one has $\varphi(F) \in \{2,4,6\}$ and $g(F) > 0.4$. For $F \ge 8$, $\varphi(F) \ge \sqrt{F/2}$ by (4.3), and $h(F) = 1.7917(\sqrt{F/2} - 0.525) - 1.2405\log F$ satisfies $h(8) > 0.06$ and is increasing for $F \ge 4$. So $g(F) > 0$ for all $F \ge 3$, and there is no solution with $p \ge 7$ and $F \ge 3$.

For $p = 3$ and $p = 5$ we use the term $c_p p^{\varphi(F)}$. A direct check shows that (5.3) holds only for $F \in \{3,4,6,8,10,12\}$ when $p = 3$ and only for $F = 6$ when $p = 5$ (for $F \ge 200$ the bound (4.3) suffices, and below 200 we use the exact $\varphi(F)$). For these F we run through the multisets S of proper divisors of F with sum F and $|S| \le \log_2((p^2-1)F/2)$, and test (5.1). The only solution is $S = \{1,1,1,1\}$ for $p = 3$, $F = 4$, that is $\{3,3,3,3\}$. All cases are listed by the programs of Appendix A and certified in Lean. □

Remark 5.2. The proof shows a little more: for an odd prime p and $k \ge 2$, the product $\prod_{f\in S}(p^f-1)$ divides p^F-1 only in the two cases of the theorem, whatever the multiset S. Taking all $f = 1$ gives the classical fact that $(p-1)^k \mid p^k-1$ with $k \ge 2$ happens only for $p = 3$ and $k \in \{2,4\}$.

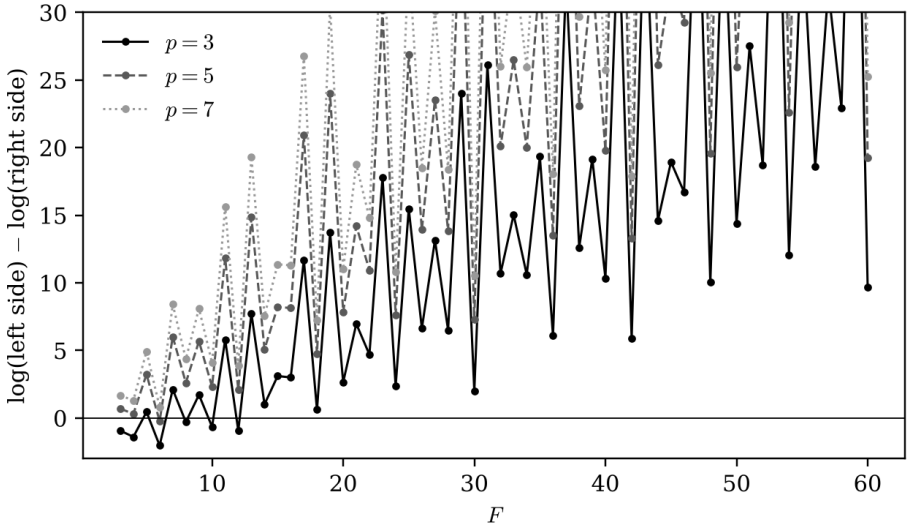


FIGURE 4. The difference of the logarithms of the two sides of (5.3) for $p = 3, 5, 7$ and $3 \le F \le 60$. Only the points below the axis survive: $F \in \{3, 4, 6, 8, 10, 12\}$ for $p = 3$ and $F = 6$ for $p = 5$. For $p \ge 7$ every point lies above the axis.

6. Consequences for rings

6.1. **Dedekind domains of positive characteristic.** If R contains a field $\mathbb{F}_p$, every residue field of R has characteristic p, so every norm multiset is a multiset of powers of p. Sections 4 and 5 give:

Theorem 6.1. *Let R be a Dedekind domain of characteristic $p > 0$ with finite residue fields.*

(a) *If $p \ge 5$, R has no Lehmer ideals and no companion ideals.*
(b) *If $p = 3$, the Lehmer ideals of R are the products of two or of four distinct primes of norm 3, and R has no companion ideals.*
(c) *If $p = 2$, the companion ideals of R are the products of t distinct primes of norm 2 and $c \ge 1$ distinct primes of norm 4 with t odd and $3^{c-1} \mid t + 2c$. A squarefree ideal with $t \le 12$ prime factors of norm 2 is a Lehmer ideal if and only if its norm multiset is in Table 1; there are none with $t = 0$, and with $t = 1$ only the products of primes of norms 2, 4 and 8.*

Proof. Lemma 2.4, Theorems 5.1, 4.7 and 4.4. □

This applies to the coordinate ring of every smooth affine curve over a finite field, and to every localization of such a ring. The number of primes of a given norm is the only arithmetic input. For $R = \mathbb{F}_q[x]$ the primes of norm q^d are the monic irreducible polynomials of degree d; there are q of degree 1 and $(q^2 - q)/2$ of degree 2.

Corollary 6.2 ([10] for $\varepsilon = -1$)**.** *Let $f \in \mathbb{F}_q[x]$ be monic and not irreducible, with $\varphi(f) > 1$.*

(a) *$\varphi(f) \mid q^{\deg f} - 1$ holds exactly in the following cases: $q = 3$ and f is a product of two distinct monic linear polynomials; $q = 2$ and f is one of*

$$x(x+1)(x^2+x+1), \quad x(x^2+x+1)(x^3+x+1), \quad x(x^2+x+1)(x^3+x^2+1),$$
$$(x+1)(x^2+x+1)(x^3+x+1), \quad (x+1)(x^2+x+1)(x^3+x^2+1).$$

(b) $\varphi(f) \mid q^{\deg f}+1$ *holds exactly when* $q=2$ *and* f *is* $x(x^2+x+1)$ *or* $(x+1)(x^2+x+1)$.

Proof. If q is a power of $p \geq 5$ there is nothing. If $q = 3^r$, Lehmer ideals need primes of norm 3, which exist only for $r = 1$; there are three monic linear polynomials, so only products of two of them occur, and no companion. If $q = 2^r$ with $r \geq 2$ there is no prime of norm 2, so $t = 0$ and there is nothing, by Theorems 4.4 and 4.7. Let $q = 2$. There are two primes of norm 2 (x and $x + 1$), one of norm 4, two of norm 8 ($x^3 + x + 1$, $x^3 + x^2 + 1$) and three of norm 16. For Lehmer ideals $t \in \{1, 2\}$; with $t = 1$ the only multiset is $\{2, 4, 8\}$, which gives the four products in the list containing a cubic, and with $t = 2$ Table 1 offers $\{2, 2, 4\}$ and $\{2, 2, 4, 4\}$, of which only the first can be realized. For companion ideals $t = 1$ and $c = 1$, since there is one prime of norm 4; the condition $3^0 \mid 3$ holds. □

Part (a) is the main theorem of Ji and Qin [10]; their list also contains the trivial polynomial $x(x + 1)$, for which $\varphi = 1$. Part (b) is the companion analogue; we have not found it in the literature.

6.2. **Number fields: ideals above one rational prime.** For a number field K and a rational prime p let $g_f(p)$ be the number of primes of $\mathcal{O}_K$ above p with residue degree f. A squarefree ideal all of whose prime factors lie above p has a norm multiset of powers of p, so the previous sections apply directly.

Theorem 6.3. *Let K be a number field.*

(a) *No ideal of $\mathcal{O}_K$ supported above a prime $p \geq 5$ is a Lehmer or a companion ideal.*
(b) *An ideal supported above 3 is a Lehmer ideal if and only if it is a product of two or of four distinct primes of degree 1 above 3. Such an ideal exists if and only if $g_1(3) \geq 2$. No ideal supported above 3 is a companion ideal.*
(c) *A companion ideal supported above 2 exists if and only if $g_1(2) \geq 1$ and $g_2(2) \geq 1$.*
(d) *If $g_1(2) \geq 1$, $g_2(2) \geq 1$ and $g_3(2) \geq 1$, or if $g_f(2) \geq 1$ for some f with $2 \leq f \leq g_1(2)$, then $\mathcal{O}_K$ has a Lehmer ideal supported above 2. The converse holds whenever $g_1(2) \leq 12$, in particular whenever $[K : \mathbb{Q}] \leq 12$.*

Proof. (a)–(c) follow from Theorems 5.1 and 4.7; in (c) the multiset $\{2, 4\}$ is companion and every companion multiset contains 2 and 4. For (d), $\{2, 4, 8\}$ is Lehmer, and if $2 \leq f \leq g_1(2)$ and $g_f(2) \geq 1$, then $\{2^{(f)}, 2^f\}$ is Lehmer by Proposition 4.5(a) and can be realized with f primes of degree 1 and one of degree f. Conversely, a Lehmer ideal above 2 has $t \leq g_1(2)$ members of norm 2; if $t \leq 12$ its multiset is in Table 1, and every entry with $t \geq 2$ contains an exponent $f \leq t$, while the only entry with $t = 1$ is $\{2, 4, 8\}$. Finally $g_1(2) \leq [K : \mathbb{Q}]$. □

Corollary 6.4. *Let K be a Galois number field. Then $\mathcal{O}_K$ has no Lehmer or companion ideal supported above a prime $p \neq 3$, and it has a Lehmer ideal supported above 3 if and only if 3 splits into at least two primes, each of residue degree 1. In a quadratic field $\mathbb{Q}(\sqrt{d})$ with d squarefree this happens exactly when $d \equiv 1 \pmod 3$, and then $3\mathcal{O}_K$ itself is the Lehmer ideal: $\varphi(3\mathcal{O}_K) = 4$ divides $\mathrm{N}(3\mathcal{O}_K) - 1 = 8$.*

Proof. In a Galois field all primes above p have the same residue degree f. Above 2, a Lehmer multiset needs members 2 and 2^f with $f \geq 2$ (Theorem 4.4: $t \geq 1$ and $S \neq \emptyset$), and a companion multiset needs both 2 and 4. Above 3 apply Theorem 6.3(b): $g_1(3) \geq 2$ means $f = 1$ and at least two primes, and in a Galois field e, f, g are constant with $efg = [K : \mathbb{Q}]$. For $K = \mathbb{Q}(\sqrt{d})$, 3 splits exactly when d is a nonzero square modulo 3. □

Corollary 6.5. *No cyclotomic field $\mathbb{Q}(\zeta_m)$ has a Lehmer or companion ideal supported above a single rational prime.*

Proof. By Corollary 6.4 only $p = 3$ with residue degree 1 and at least two primes could work. Write $m = 3^a m'$ with $3 \nmid m'$. The residue degree of 3 is the order of 3 modulo m', which is 1 only when $m' \in \{1, 2\}$; then $\mathbb{Q}(\zeta_m) = \mathbb{Q}(\zeta_{3^a})$ and 3 is totally ramified, with a single prime above it. □

Example 6.6. In $\mathbb{Q}(\sqrt{-2})$ the ideal $(3) = (1 + \sqrt{-2})(1 - \sqrt{-2})$ is a Lehmer ideal with $M = 2$. In the quartic field $K = \mathbb{Q}(\alpha)$ with $\alpha^4 + \alpha + 2 = 0$ (discriminant $2021 = 43 \cdot 47$) the prime 2 factors as $\mathfrak{p}_1\mathfrak{p}_2\mathfrak{q}$ with residue degrees $1, 1, 2$, by Dedekind's criterion, because $x^4 + x + 2 \equiv x(x + 1)(x^2 + x + 1) \pmod 2$. The ideal $\mathfrak{p}_1\mathfrak{p}_2\mathfrak{q} = 2\mathcal{O}_K$ has multiset $\{2, 2, 4\}$, so $\varphi(2\mathcal{O}_K) = 3$ divides $\mathrm{N}(2\mathcal{O}_K) - 1 = 15$, and the ideals $\mathfrak{p}_1\mathfrak{q}$ and $\mathfrak{p}_2\mathfrak{q}$ are companion ideals with $\varphi = 3$ and $\mathrm{N} + 1 = 9$.

7. Odd multisets with several primes

The remaining admissible multisets consist of odd prime powers belonging to at least two different primes. Here no classification is in sight: over $\mathbb{Z}$ this contains Lehmer's original problem. We give the complete lists for few members, in the four arithmetic settings of Section 3.2, and describe how the solutions are related.

7.1. **All number fields.**

Theorem 7.1. (a) *The Lehmer multisets of odd prime powers with at most* 7 *members are*

$$\{3, 3\}, \quad \{3, 3, 3, 3\}, \quad \{5, 5, 9, 9, 89\}, \quad \{3, 3, 5, 5, 9, 89\}, \quad \{3, 3, 3, 3, 5, 5, 89\},$$

with $M = 2, 5, 2, 4, 8$ *respectively.*

(b) *The companion multisets of odd prime powers with at most* 6 *members are the twelve multisets of Table 2.*

Proof. By Theorem 3.3 and Proposition 3.2 the search tree of Section 3.2 in mode U is finite; it was traversed completely, for every M allowed by Lemma 3.1, by the program described in Appendix A. The sizes of the trees are in Table 4. That the listed multisets are admissible is checked by hand below and certified in Lean. □

The three Lehmer multisets with several primes have the same norm,

$$\mathrm{N} = 3^4 \cdot 5^2 \cdot 89 = 180225, \qquad \mathrm{N} - 1 = 2^{14} \cdot 11,$$

with $\varphi = 4^2 \cdot 8^2 \cdot 88 = 90112 = 2^{13} \cdot 11$ for $\{5, 5, 9, 9, 89\}$. The independence condition holds: $5 \nmid 8, 88$; $3 \nmid 4, 88$; $89 \nmid 4, 8$. The other two arise from the first by the following operation.

Lemma 7.2. *Let* $\mathcal{Q}$ *be an odd* ε*-admissible multiset that contains* 9*. Replacing one member* 9 *by two members* 3 *gives an* ε*-admissible multiset with the same norm and twice the value of* M*.*

Proof. The norm is unchanged and φ is divided by $(9 - 1)/(3 - 1)^2 = 2$, so φ still divides $\mathrm{N} + \varepsilon$ and M doubles. □

The lemma applied twice to $\{5, 5, 9, 9, 89\}$ gives the other two Lehmer multisets, and applied to $\{3, 9, 9, 23, 131\}$ it gives the companion multiset $\{3, 3, 3, 9, 23, 131\}$. The operation cannot be reversed in general: $\{3, 3, 3, 3\}$ would come from $\{3, 3, 9\}$, which is not admissible.

TABLE 2. Companion multisets of odd prime powers with at most six members. The column "Fields" says in which settings the multiset can occur: $\mathbb{Z}$ (distinct primes), Q (quadratic fields), G ($\mathbb{Z}[i]$), E ($\mathbb{Z}[\omega]$), listing the settings whose residue rules (Section 3) the multiset satisfies; a quadratic entry is realised in infinitely many fields $\mathbb{Q}(\sqrt{d})$, and every multiset occurs in infinitely many number fields.

$\mathcal{Q}$	$\mathrm{N}(\mathcal{Q})$	M	Fields
$\{3,5\}$	15	2	$\mathbb{Z}$, Q
$\{3,5,17\}$	255	2	$\mathbb{Z}$, Q
$\{5,5,5,43\}$	5375	2	–
$\{3,5,17,257\}$	65535	2	$\mathbb{Z}$, Q
$\{3,9,9,23,131\}$	732159	2	–
$\{3,5,17,353,929\}$	83623935	2	$\mathbb{Z}$, Q
$\{3,5,17,257,65537\}$	4294967295	2	$\mathbb{Z}$, Q
$\{3,3,3,9,23,131\}$	732159	4	–
$\{5,7,7,13,13,619\}$	25629695	2	Q
$\{3,11,11,11,677,3659\}$	9891231999	2	–
$\{5,5,5,43,5413,786349\}$	22878725861375	2	–
$\{3,5,17,353,929,83623937\}$	6992962672132095	2	$\mathbb{Z}$, Q

7.2. **A biquadratic Lehmer ideal.** Corollary 2.7 says that each multiset of Theorem 7.1(a) is the norm multiset of a Lehmer ideal in infinitely many number fields. For $\{5,5,9,9,89\}$ a Galois field of degree four suffices.

Proposition 7.3. *Let $K=\mathbb{Q}(i,\sqrt{89})$. Then 5 splits completely in K, 3 has residue degree 2 and splits into two primes, and 89 ramifies in $\mathbb{Q}(\sqrt{89})$ and splits in $\mathbb{Q}(i)$, so $89\mathcal{O}_K=\mathfrak{q}_1^2\mathfrak{q}_2^2$ with $\mathrm{N}(\mathfrak{q}_j)=89$. If $\mathfrak{p}_1,\mathfrak{p}_2$ are two of the four primes above 5 and $\mathfrak{a}_1,\mathfrak{a}_2$ are the two primes above 3, the ideal*

$$I=\mathfrak{p}_1\mathfrak{p}_2\mathfrak{a}_1\mathfrak{a}_2\mathfrak{q}_1$$

is a Lehmer ideal of $\mathcal{O}_K$ with $\mathrm{N}(I)=180225$, $\varphi(I)=90112$ and $M=2$.

Proof. The three quadratic subfields are $\mathbb{Q}(i)$, $\mathbb{Q}(\sqrt{89})$ and $\mathbb{Q}(\sqrt{-89})$. Since $5\equiv 1 \pmod 4$ and $89\equiv 4 \pmod 5$ is a square, 5 splits in $\mathbb{Q}(i)$ and in $\mathbb{Q}(\sqrt{89})$, hence completely in K. The prime 3 is inert in $\mathbb{Q}(i)$ and in $\mathbb{Q}(\sqrt{89})$ ($89\equiv 2 \pmod 3$), and splits in $\mathbb{Q}(\sqrt{-89})$ ($-89\equiv 1 \pmod 3$), so its decomposition group has order 2 and $3\mathcal{O}_K=\mathfrak{a}_1\mathfrak{a}_2$ with $\mathrm{N}(\mathfrak{a}_j)=9$. The prime 89 ramifies in $\mathbb{Q}(\sqrt{89})$ and splits in $\mathbb{Q}(i)$ because $89\equiv 1 \pmod 4$. Hence $\mathcal{Q}(I)=\{5,5,9,9,89\}$, and Lemma 2.4 applies. The factorizations were also confirmed with PARI/GP. □

The multisets $\{3,3,5,5,9,89\}$ and $\{3,3,3,3,5,5,89\}$ contain primes of residue degrees 1 and 2 above 3, so they cannot occur in a Galois field; by Theorem 2.5 they occur in fields of every degree $n\geq 4$.

7.3. **Quadratic fields, Gaussian and Eisenstein integers.** In a quadratic field a rational prime contributes $\{p,p\}$ (split), $\{p\}$ (split or ramified, one prime chosen) or $\{p^2\}$ (inert), so the admissible multisets that can occur are those of mode Q. By Lemma 2.2(c) and the discussion of Section 4, an even Lehmer multiset in a quadratic field would contain 2 and some 2^f with $f\geq 2$, which is impossible because 2 contributes either $\{2\}$, $\{2,2\}$ or $\{4\}$; even companion multisets need 2 and 4 and are impossible for the same reason.

Theorem 7.4. *Let K be a quadratic field.*

(a) *A Lehmer ideal of $\mathcal{O}_K$ with at most 7 prime factors is $3\mathcal{O}_K$, and it is a Lehmer ideal exactly when 3 splits in K.*

(b) *In $\mathbb{Z}[i]$ there is no Lehmer ideal with at most 10 prime factors, and in $\mathbb{Z}[\omega]$, $\omega = e^{2\pi i/3}$, there is none with at most 14 prime factors.*

Proof. The searches in modes Q, G and E (Table 4) find, apart from $\{3,3\}$ in mode Q, no odd Lehmer multiset; in modes G and E the multiset $\{3,3\}$ is excluded because 3 is inert in $\mathbb{Z}[i]$ and ramified in $\mathbb{Z}[\omega]$. The even case was excluded above. □

Since $\mathbb{Z}[i]$ and $\mathbb{Z}[\omega]$ are principal ideal domains, part (b) says: there is no Gaussian integer α, not a unit times a prime, with at most 10 prime factors counted without multiplicity and $|(\mathbb{Z}[i]/\alpha)^\times|$ dividing $\mathrm{N}(\alpha)-1$; similarly for Eisenstein integers.

The companion condition behaves differently. In every quadratic field in which neither 3 nor 5 is inert, a product of a prime above 3 and a prime above 5 is a companion ideal, with multiset $\{3,5\}$ as for the integer 15; and there are quadratic companion ideals with no counterpart in $\mathbb{Z}$, for example those with multiset $\{5,7,7,13,13,619\}$, which occurs in every quadratic field in which 7 and 13 split and 5, 619 do not stay inert.

Proposition 7.5. *The odd companion multisets of mode Q with at most seven members are the six sets of Theorem 8.1(b), the multiset $\{5,7,7,13,13,619\}$ and*

$$\{5,7,7,13,13,619,25629697\},$$

all with $M=2$. In modes G and E there is no companion multiset with at most seven members.

Proof. Complete traversal of the trees in modes Q, G and E for $\varepsilon=+1$ (Table 4). For the new multiset, 25629697 is prime, $25629697-1 = 2^{10}\cdot 3^5\cdot 103$, and $25629697 = 5\cdot 7^2\cdot 13^2\cdot 619+2$, so with $\mathrm{N}' = 5\cdot 7^2\cdot 13^2\cdot 619$ and $\varphi' = \mathrm{N}'/2+1/2$ one has $\mathrm{N}+1 = (\mathrm{N}'+1)^2 = 4\varphi'^2$ and $\varphi = \varphi'(\mathrm{N}'+1) = 2\varphi'^2$. □

The new multiset arises from $\{5,7,7,13,13,619\}$ in the same way as 83623937 arises from $3\cdot 5\cdot 17\cdot 353\cdot 929$: if $\mathcal{Q}$ is companion with $M=2$ and $q=\mathrm{N}(\mathcal{Q})+2$ is a prime power, then $\mathrm{N}(\mathcal{Q})q+1 = (\mathrm{N}(\mathcal{Q})+1)^2$ and $\varphi(\mathcal{Q})(q-1) = (\mathrm{N}(\mathcal{Q})+1)^2/2$, so appending q gives a companion multiset with $M=2$.

8. The rational integers

For $R=\mathbb{Z}$ the multisets are sets of distinct primes, and Lehmer's question is whether $\varphi(n)\mid n-1$ forces n to be prime. Lehmer proved that a composite solution is odd, squarefree and has at least seven prime factors [12]; Cohen and Hagis raised this to fourteen [7], and a computation of Renze [16] to fifteen. Pomerance [15] showed $n<k^{2^k}$ for a solution with k prime factors, and Burek and Żmija improved this bound [5]. On the analytic side, the number of composite solutions up to x is at most of order $x^{1/2}$ up to logarithmic factors [3], and there are results for special forms of n, such as repunits [6]. The weaker conditions $\varphi(n)\mid(n-1)^k$ define the k-Lehmer numbers [8, 13], which are related to Carmichael numbers [1].

Mode Z of our search gives an independent check of the bound $\omega(n)\ge 15$ with a program written from the description in Section 3.2.

Theorem 8.1. (a) *There is no composite n with $\varphi(n)\mid n-1$ and at most 14 prime factors.*

(b) *The composite n with $\varphi(n)\mid n+1$ and at most 7 prime factors are*

$3\cdot 5$, $3\cdot 5\cdot 17$, $3\cdot 5\cdot 17\cdot 257$, $3\cdot 5\cdot 17\cdot 257\cdot 65537$, $3\cdot 5\cdot 17\cdot 353\cdot 929$, $3\cdot 5\cdot 17\cdot 353\cdot 929\cdot 83623937$,

all with $M=2$.

Proof. Complete traversal of the trees of Section 3.2 in mode `Z`; the sizes are in Table 4. In (a) the tree for $k = 14$ has 32489 nodes, of which 29631 are at depth 12, where the last two primes are determined by the forced value; about $1.6 \cdot 10^7$ candidates for q_{13} are tested in total, in about two minutes on one core. □

Part (a) does not improve the known bound; its interest is that it is obtained by the same short program that handles all other rings in this paper, so that the integer case serves as a check of the program. The companion solutions in (b) consist of a chain of Fermat primes followed by primes q with $q-1$ a power of two times small factors; for instance in $3 \cdot 5 \cdot 17 \cdot 353 \cdot 929$ one has $353-1 = 2^5 \cdot 11$ and $929-1 = 2^5 \cdot 29$, and the next member $83623937 = 3 \cdot 5 \cdot 17 \cdot 353 \cdot 929+2$ is again the forced last member $(MB+1)/C$ of Proposition 3.2.

Remark 8.2. The product formula also shows that a small Lehmer number must have $M = 2$ and $3 \nmid n$. Write $\rho(n) = n/\varphi(n) = M + 1/\varphi(n)$. If $3 \mid n$, the other prime factors are $\equiv 2 \pmod 3$ and $M \equiv 1 \pmod 3$ (Section 3.2), so $M \geq 4$ and $\rho(n) > 4$. If a set of r primes from a given list has $\rho > \rho_0$, then the r smallest primes of the list have larger ρ and smaller product. The 1538 smallest primes $p \equiv 2 \pmod 3$ together with 3 give $\rho \leq 4$, so $3 \mid n$ forces $n > 10^{6082}$. If $3 \nmid n$ and $M \geq 3$, then $\rho(n) > 3$, and the 32 smallest primes $p \geq 5$ give $\rho \leq 3$, so $n > 10^{56}$. Hence every Lehmer number below 10^{56} satisfies $n - 1 = 2\varphi(n)$ and has all prime factors at least 5. Such an n has at least 15 prime factors by Theorem 8.1(a), and at most 23 if $n < 10^{37}$, because the 24 smallest primes $p \geq 5$ have product larger than 10^{37}. For 16 to 23 prime factors, an exhaustive search of the case $M = 2$ below 10^{37} by the author (program `tree_below.c` with its complete output in the directory `below` of `https://github.com/creelie/eLhmr`, about $2.3 \cdot 10^{10}$ nodes) found no solution. Hence a composite $n < 10^{37}$ with $\varphi(n) \mid n-1$ would have exactly 15 prime factors, all at least 5, and satisfy $n - 1 = 2\varphi(n)$.

In the language of ideals, the integer problem asks for ε-admissible sets of distinct primes. Theorem 7.1 shows that once repetition and prime powers are allowed, Lehmer multisets with several primes appear at five members already. The smallest one, $\{5, 5, 9, 9, 89\}$, needs two primes of norm 5 and two of norm 9; this is the kind of structure that $\mathbb{Z}$ lacks.

9. Status of related conjectures and questions

Table 3 collects the statements related to Lehmer's problem that this paper touches, with their status after the results above. "Open" means that we neither prove nor disprove the statement; the comment says what is added here.

We close with three questions suggested by the computations.

Question 9.1. *Is $3\mathcal{O}_K$ the only Lehmer ideal in a quadratic field K? Is there any Lehmer element in $\mathbb{Z}[i]$ or in $\mathbb{Z}[\omega]$?*

Question 9.2. *Are there infinitely many Lehmer multisets of odd prime powers that involve at least two primes? Is there one whose norm differs from* 180225*?*

Question 9.3. *For the multisets of powers of two, is $\max\{F\}$ over the Lehmer multisets with t members 2 equal to $O(t)$? In Table 1 it lies between $2t$ and $6t$ for $2 \leq t \leq 12$.*

For Question 9.3, Proposition 4.5(a) with $f = t$ gives $\max F \geq 2t$, and the proof of Theorem 4.3 gives $\varphi(F) \leq t + O(\log F)$, hence $\max F = O(t \log\log t)$ because $\varphi(F) \gg F/\log\log F$.

TABLE 3. Conjectures and questions related to Lehmer's totient problem.

Statement	Status	Comment
Lehmer: $\varphi(n) \mid n-1$ implies n prime [12]	open	$\omega(n) \geq 15$ rechecked by an independent program (Theorem 8.1); a solution below 10^{37} would have exactly 15 prime factors (Remark 8.2).
Companion over $\mathbb{Z}$: the six known composite solutions of $\varphi(n) \mid n+1$ are all	open	no other with at most 7 prime factors (Theorem 8.1).
Lehmer over $\mathbb{F}_q[x]$ [10]	proved [10]	new proof from Theorem 6.1; companion analogue proved (Corollary 6.2).
Lehmer in every Dedekind domain of characteristic $p \geq 3$	proved	complete answer, Theorem 6.1(a),(b).
Lehmer in characteristic 2	partly proved	finite for each t; complete for $t \leq 12$ (Theorem 4.4); companion complete (Theorem 4.7).
Bound $n < k^{2^k}$ for k prime factors [15]	extended	uniform bound $(k2^{k+1})^{2^k-1}$ for all Dedekind domains (Theorem 3.3).
Lehmer ideals in number fields exist	proved	$3\mathcal{O}_K$ in $\mathbb{Q}(\sqrt{-2})$; every admissible multiset occurs (Corollary 2.7).
Lehmer ideals above one rational prime	proved	complete for p odd and for companion; for $p = 2$ when $g_1(2) \leq 12$ (Theorem 6.3).
Lehmer ideals in Galois and cyclotomic fields above one prime	proved	Corollaries 6.4 and 6.5.
Lehmer elements of $\mathbb{Z}[i]$ and $\mathbb{Z}[\omega]$ do not exist	open	none with at most 10, resp. 14, prime factors (Theorem 7.4).
In quadratic fields only $3\mathcal{O}_K$ is Lehmer	open	true up to 7 prime factors (Theorem 7.4).
Question 4.6 (an exponent $f \leq t$)	open	true for $t \leq 12$.
Infinitely many odd Lehmer multisets with several primes	open	five with at most 7 members (Theorem 7.1).

APPENDIX A. ALGORITHMS, PROGRAMS AND TREE SIZES

A.1. **The programs.** All finite statements were checked by two programs written independently from the descriptions in Sections 3.2, 4 and 5.

(1) A C program using GMP traverses the search tree of Section 3.2 for a given mode, sign ε and number of members k, looping over all M with $2 \leq M \leq \lfloor (3/2)^k \rfloor + 1$; this is the bound of Lemma 3.1 with 2 replaced by $3/2$, since $q/(q-1) \leq 3/2$ for odd q. Intervals of candidate prime powers are produced by a segmented sieve with base primes up to 10^7 and a sorted list of all prime powers $p^e \leq 10^{14}$ with $e \geq 2$; primality above 10^{14} is decided by a Miller–Rabin test with the twelve prime bases up to 37, which is deterministic for all 64-bit integers. A forced last member beyond 2^{64} is tested with the probabilistic test of GMP, which never rejects a prime, so no solution can be lost there; every reported solution was confirmed afterwards by exact factorization. The program counts every node at which an interval would have to start at or beyond 2^{62} while the reach condition still holds, and every interval whose right end would exceed 2^{62}; both counters are zero in all runs reported here, so no branch was cut.

(2) A Julia program traverses the same tree with exact integers by a separate implementation (without the modulo-3 rule of mode Z), and also performs the enumerations of Theorems 4.4 and 5.1, the analytic checks of the sets E_t and of the functions g and h in the proof of Theorem 5.1, and a brute-force test of all multisets of powers of an odd prime $p < 200$ with $F \le 16$. For $k \le 6$ the node counts of the two traversals coincide in modes U, Q, G and E, and the solution lists coincide in all modes.
(3) The Python program used to discover the binary classification agrees with both.

A.2. **Tree sizes.** Table 4 gives, for each mode and number k of members, the number of nodes of the Lehmer search tree, the number of nodes at depth $k-2$ (where the last member is forced), and the number of candidates q_{k-1} tested there, summed over M. A node is counted once for each value of M. Figure 5 shows the growth.

Table 4: Sizes of the search trees for each mode, sign ε and number k of members: nodes, nodes at depth $k-2$, candidates for q_{k-1}, and solutions found.

Mode	ε	k	nodes	depth $k-2$	candidates	solutions
Z	−1	2	2	1	1	0
Z	−1	3	4	0	0	0
Z	−1	4	9	0	0	0
Z	−1	5	15	0	0	0
Z	−1	6	24	0	0	0
Z	−1	7	39	0	0	0
Z	−1	8	64	1	0	0
Z	−1	9	101	1	0	0
Z	−1	10	147	1	0	0
Z	−1	11	231	7	17	0
Z	−1	12	426	63	217	0
Z	−1	13	1 446	730	11 936	0
Z	−1	14	32 489	29 631	15 922 545	0
U	−1	2	2	1	1	1
U	−1	3	4	1	1	0
U	−1	4	14	5	10	1
U	−1	5	44	19	54	1
U	−1	6	193	122	4 350	1
U	−1	7	9 242	8 929	114 411 239	1
Q	−1	2	2	1	1	1
Q	−1	3	4	1	1	0
Q	−1	4	13	4	4	0
Q	−1	5	32	8	27	0
Q	−1	6	116	63	3 409	0
Q	−1	7	7 159	6 976	112 856 127	0
G	−1	2	2	1	0	0
G	−1	3	3	0	0	0
G	−1	4	7	1	0	0
G	−1	5	13	1	0	0

Mode	ε	k	nodes	depth $k-2$	candidates	solutions
G	-1	6	22	3	4	0
G	-1	7	41	6	11	0
G	-1	8	87	29	120	0
G	-1	9	402	270	7 078	0
G	-1	10	16 230	15 587	20 159 732	0
E	-1	2	2	1	1	0
E	-1	3	4	1	0	0
E	-1	4	8	0	0	0
E	-1	5	13	0	0	0
E	-1	6	20	0	0	0
E	-1	7	32	0	0	0
E	-1	8	49	0	0	0
E	-1	9	80	1	0	0
E	-1	10	119	2	2	0
E	-1	11	195	4	7	0
E	-1	12	332	32	104	0
E	-1	13	856	316	2 943	0
E	-1	14	8 517	7 024	1 655 269	0
Z	$+1$	2	2	1	1	1
Z	$+1$	3	4	1	1	1
Z	$+1$	4	10	1	3	1
Z	$+1$	5	20	3	22	2
Z	$+1$	6	77	44	3 344	1
Z	$+1$	7	6 943	6 826	112 850 405	0
U	$+1$	2	2	1	1	1
U	$+1$	3	4	1	1	1
U	$+1$	4	14	5	10	2
U	$+1$	5	44	19	54	3
U	$+1$	6	193	122	4 350	5
Q	$+1$	2	2	1	1	1
Q	$+1$	3	4	1	1	1
Q	$+1$	4	13	4	4	1
Q	$+1$	5	32	8	27	2
Q	$+1$	6	116	63	3 409	2
Q	$+1$	7	7 159	6 976	112 856 127	1
G	$+1$	2	2	1	0	0
G	$+1$	3	3	0	0	0
G	$+1$	4	7	1	0	0
G	$+1$	5	13	1	0	0
G	$+1$	6	22	3	4	0
G	$+1$	7	41	6	11	0
E	$+1$	2	2	1	1	0
E	$+1$	3	4	1	0	0
E	$+1$	4	8	0	0	0

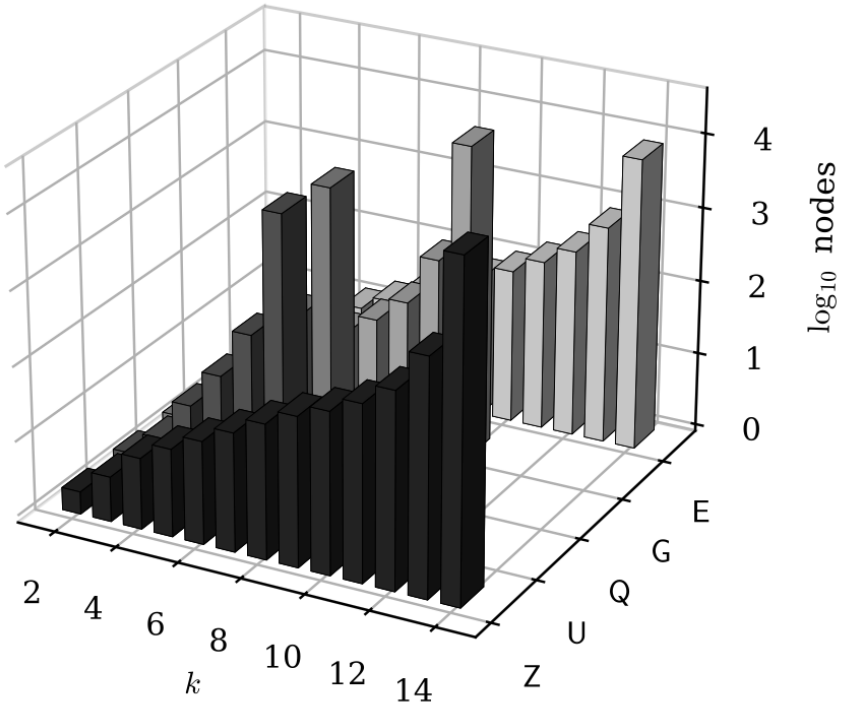


FIGURE 5. Number of nodes of the Lehmer search trees, on a logarithmic scale, for the five modes and the values of k in Table 4.

Mode	ε	k	nodes	depth $k-2$	candidates	solutions
E	$+1$	5	13	0	0	0
E	$+1$	6	20	0	0	0
E	$+1$	7	32	0	0	0

A.3. **Two checks by hand.** Each listed multiset is admissible by a short computation. For $\mathcal{Q} = \{5, 5, 9, 9, 89\}$,

$$\mathrm{N}(\mathcal{Q}) = 5^2 \cdot 3^4 \cdot 89 = 180225, \qquad \varphi(\mathcal{Q}) = 4^2 \cdot 8^2 \cdot 88 = 90112,$$

so that $\mathrm{N}(\mathcal{Q})-1 = 2\varphi(\mathcal{Q})$ and $M = 2$. For the companion multiset $\mathcal{Q} = \{3, 11, 11, 11, 677, 3659\}$ one finds

$$\mathrm{N}(\mathcal{Q}) = 3 \cdot 11^3 \cdot 677 \cdot 3659 = 9891231999,$$

$$\varphi(\mathcal{Q}) = 2 \cdot 10^3 \cdot 676 \cdot 3658 = 4945616000,$$

and $\mathrm{N}(\mathcal{Q})+1 = 9891232000 = 2\varphi(\mathcal{Q})$. In both cases the independence conditions of Lemma 2.2 hold: no prime of $\mathcal{Q}$ divides $q-1$ for a member q, as the factorizations of the numbers $q-1$ show.

Appendix B. The Lean certificate

The file `LehmerIdeals.lean` uses only the core library of Lean 4 (version 4.24.0) and contains 381 theorems. Each is a closed equation between Boolean, numeral or list expressions and is proved by `decide +kernel`, which makes the kernel evaluate both sides; there is no `sorry` and no `native_decide`, and `#print axioms` reports that no theorem depends on any axiom. The whole file is checked in about half a minute. In the binary enumeration, a partial multiset S is extended by f only if $\prod(2^{f'}-1)$ over the extended multiset still divides 2^F-1; since a product divides 2^F-1 only if every partial product does, this pruning loses no solution, and it keeps the kernel computation small. It certifies:

(1) for each $t \le 12$ and each $F \in E_t$, the complete list of multisets S of proper divisors $f \ge 2$ of F with $\sum f = F - t$ satisfying (4.1), which together are the rows of Table 1 (370 theorems `bin_t`t`_F`F); the sets E_t themselves are computed in the Julia program from the analytic inequality of Theorem 4.3;
(2) the 3-adic criteria of Proposition 4.5(b) and Theorem 4.7 for $t \le 40$, $c \le 7$, and the non-divisibility $2^f - 1 \nmid 2^F + 1$ for $3 \le f \le 30$, $F < 120$;
(3) the finite parts of Theorem 5.1 for $p = 3$ and $p = 5$;
(4) the admissibility, with the stated M and the independence condition, of the five Lehmer multisets of Theorem 7.1(a), the twelve companion multisets of Table 2 and the seven-member multiset of Proposition 7.5;
(5) the congruences used in Proposition 7.3, Example 6.6 and Corollary 6.2.

What the certificate does not cover are the analytic inequalities (the definition of E_t, the bounds in the proof of Theorem 5.1) and the traversals of Section 7, which are checked by the two programs.

Author contributions

Deep Bhattacharjee is the sole author. He posed the problems in the form treated here, proved the results, wrote the paper, drew the figures, and wrote the C, Julia and Lean 4 programs.

Open Gluon and Anaxia

Two tools built by the author organised this project. Open Gluon is a workspace that keeps pinned copies of the mathematical libraries, the source record of the project and its checks in one reproducible place. Anaxia, the engine of the eBLLM project, stores each construction as a route with its input, its output, the obligations that must be discharged and a test that makes the route fail when an obligation is skipped, and it certifies finite statements in exact integer arithmetic. It was used here to keep the search trees of Section 7, their overflow counters and their two independent traversals in one record. Every statement of this paper is proved in the text.

Declarations

Data and code availability. The C, Julia and Lean 4 programs, the search logs and the figure scripts are archived at [Zenodo DOI to be added on release].

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